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Prof. Dr. Lincy Pyl and Selection Committee

Department of Mechanics of Materials and Constructions (MeMC)

Vrije Universiteit Brussel

Subject: Motivation letter – PhD candidate, Mechanics of Materials and Constructions (M²S)

Introduction video: <https://drive.google.com/file/d/14s-V07z3fopEllnUP8vmMJPMxQW57mg9/view>

Dear Prof. Pyl and Selection Committee,

I am applying for the PhD position in Mechanics of Materials and Constructions at VUB. I am keen to join the Mechanics of Advanced Materials and Structures (M²S) group and to develop, together with my supervisors, a doctoral project on advanced characterisation and mechanics-driven design of high-performance and sustainable materials and structures. I want to complete a focused four-year PhD, publish, present internationally, and contribute to teaching and departmental life on the Elsene campus.

I am especially motivated by experimental work on construction materials. During my Diploma and Bachelor's in Civil Engineering in India I completed laboratory courses on concrete, aggregates, steel, soil and water. I carried out practical tests such as slump and cube compressive strength (IS 1199, IS 516), aggregate and sand grading (IS 2386), steel tensile testing (IS 1608 / IS 1786), and Proctor compaction and Atterberg limits (IS 2720). Those labs stayed with me. They showed that sustainable construction depends on how materials perform under load, how we characterise them, and how we design structures that use materials well. I want to return to that foundation at research level inside MeMC.

My Master's in Environmental Engineering at HTW Dresden builds on this civil base. I took Construction Materials (B663), Geotechnical Project (B634), and Water, Air and Soil Analysis (B626) with laboratory practicals, as well as Soil Management, Water and Wastewater Treatment, and Solid Waste Management and Recycling. My master's thesis, *Agile Project Management in the Construction Industry* (grade 2,5), studied how construction teams can manage risk and use resources more carefully. After that work I became more interested in materials performance and in how mechanical behaviour and sustainability can be designed together. M²S themes (characterisation and monitoring, high-performance and sustainable materials and structures, and adaptive material systems) are the direction I want to take.

Professionally I have worked in Belgium and want to build my research career there. At iosting I worked with a construction partner on bringing circular practices into project delivery, including waste-management workflows. That role taught me to work with industry partners and to connect civil practice with clear process thinking. I am now ready to deepen the materials and mechanics side of my profile through doctoral research.

I am an early-career researcher with limited paid scientific research experience, and I am ready to prepare applications for external fellowships such as FWO. I work in a calm and structured way, independently and in a team. I welcome teaching of exercise and laboratory sessions, Open Days and departmental tasks. I speak English fluently (C1). I have basic Dutch (A1/A2) from my time in Belgium and I am committed to improving it for Dutch-language teaching where needed. I am based in Pune and open to relocating to Brussels for a start on 1 October 2026.

I would be glad to shape a PhD topic with MeMC supervisors around mechanical characterisation of construction materials with a sustainability or reuse angle, or mechanics-driven design of more resource-efficient structural systems, always grounded in experiments and clear engineering questions. I am motivated to learn advanced characterisation methods, strengthen my analytical and numerical skills, and contribute high-quality publications from M²S.

Thank you for considering my application. I look forward to the possibility of an interview.

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Yours sincerely,

Pooja Awasare